

Index

National Artificial Intelligence



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1. Document Purpose

This document aims to introduce the National AI Index (NAII) by outlining its objectives and the methodology adopted in its development. It also covers the key components on which the Index relies to assess the readiness of government entities to adopt AI technologies. The document presents the maturity levels for AI adoption by government entities, providing a detailed description of each level, then explains the evaluation questions included in the Index, and concludes with the mechanism for analyzing results.

2. Index Overview

The National AI Index (NAII) aims to assess the readiness level of government entities to adopt AI technologies, monitor their progress on a periodic basis, and provide practical recommendations on strengths and potential areas for improvement for each entity, thereby supporting the development and use of these technologies at the national level. The Index consists of three main pillars, seven sub-pillars, and twenty-three domains to ensure a comprehensive measurement process.

The design of the Index considers current challenges and future opportunities faced by government entities in adopting AI. The most notable of these challenges include weak alignment between AI initiatives and national priorities, limited availability of suitable data for training, variations in the readiness of technical infrastructure across entities, a shortage of specialized national talent, and a lack of unified and clear measurement standards.

Index Objectives:

- **Aligning** government efforts and priorities in the field of AI.
- **Providing and improving** the necessary enablers for AI adoption within government entities.
- **Enabling** the launch and adoption of AI products in key sectors and ensuring impact realization.

The Index delivers results that reflect each entity's adoption level while also providing the necessary support and recommendations to strengthen their capabilities in developing effective and sustainable AI solutions and applications that directly contribute to achieving the objectives of Saudi Vision 2030. Additionally, the Index aims to empower government entities to develop AI products in priority sectors, provide the essential enablers to support the sustainability of these efforts, and maximize their impact.

3. Index Development Methodology

The development of national indices is one key tool for strengthening the governance of emerging technologies and guiding efforts toward achieving strategic objectives. With the rapid expansion in AI adoption across both the public and private sectors, there is a clear need to design a national index that reflects entities' readiness to adopt AI and identifies opportunities for advancement in this vital field.

The methodology for developing the Index is based on three interconnected stages aimed at achieving comprehensiveness and accuracy in building the Index, including surveying and analyzing benchmark studies, designing the Index by defining its key components and measurement mechanism, and performing measurement procedures. This process includes conducting workshops, completing and

verifying questionnaires, and finally analyzing the results to determine each entity’s level of adoption. The findings are then compiled into reports that support decision-makers in enhancing AI utilization and improving enterprise performance.

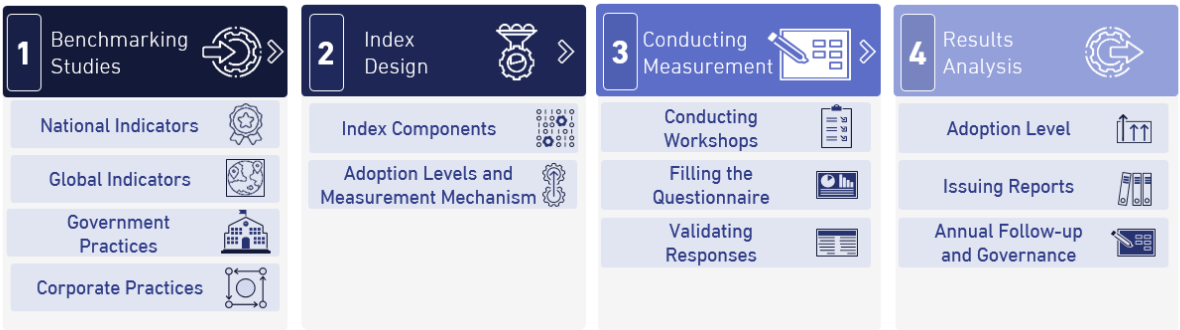


Figure (1): Index Development Methodology

3.1 Analysis

The development of the National AI Index was based on an in-depth analysis of a comprehensive set of national and international references. This analysis included studying national indices in Saudi Arabia and reviewing regulatory frameworks related to AI. It also examined international AI indices, in addition to analyzing the experiences of governments and corporations in this field.

3.2 Design

The design phase of the National AI Index aims to build a comprehensive and flexible measurement tool that reflects the reality of entities in terms of strategic direction, internal readiness, and achieved outcomes and impact. The design process considered alignment with the local context and leveraged global best practices to ensure the reliability of results and their applicability across various sectors.

3.2.1 Index Framework

The Index consists of three main pillars: Directions, Enablers, and Outputs. These pillars include seven main sub-pillars, divided into 23 sub-domains covering strategic, regulatory, technical, operational, and human aspects. These components were carefully selected to provide a comprehensive and integrated view of entities' readiness levels in the field of AI.

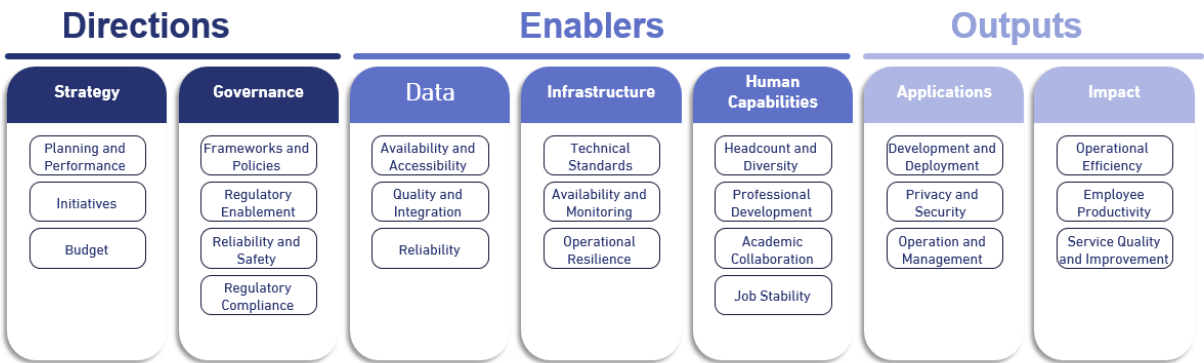


Figure (2): Index Framework

A. Directions:

Strategy

Leading entities adopt ambitious strategies to deeply integrate AI into their operations, with the aim of enhancing operational efficiency, reducing costs, and delivering personalized services. These entities focus on leveraging AI to achieve a competitive advantage, foster innovation, and improve service quality through advanced analytics.

Such strategies outline how the organization harnesses AI to achieve its objectives, defining priorities, requirements, actions, timelines, and the governance framework necessary for sustainable implementation. This sub-pillar covers:

- **Planning and Performance**

Formulating the strategy and vision serves as the enterprise reference for directing AI efforts toward measurable objectives and identifying priority domains. This contributes to building a shared understanding, linking initiatives to enterprise objectives, and ensuring alignment with national strategies and Saudi Vision 2030.

- **Initiatives**

Initiatives serve as a tool of translating the strategic vision into practical projects that deliver tangible impact. It includes operational and experimental initiatives aimed at improving performance, such as process automation and the delivery of intelligent recommendations.

- **Budget**

This refers to the dedicated budget for implementing AI initiatives within the entity, which is essential for turning plans into outcomes. It reflects a strategic commitment in supporting digital transformation.

Governance

As AI adoption develops, governance becomes a necessity to ensure the responsible and safe use of technologies. Governance frameworks define roles, responsibilities, and decision-making mechanisms, striking a balance between innovation, transparency, and compliance, thereby enhancing public trust and achieving enterprise excellence. AI governance encompasses a set of frameworks, policies, and enterprise structures that regulate the development and use of AI to ensure compliance and promote safe, responsible use through periodic reviews and the adoption of leading local and global practices. This includes:

- **Frameworks and Policies**

Regulatory frameworks form the foundation for effective governance. They support the organized use of smart technologies, strengthen public trust, and consider cultural and developmental specificities.

- **Regulatory Enablement**

This includes establishing a regulatory unit dedicated to AI within the entity to strengthen governance and enterprise transformation, along with clearly defining its mandates.

- **Reliability and Safety**

This involves ensuring the accuracy and stability of AI systems through control procedures, adherence to national standards, and mitigation of potential risks, in alignment with relevant regulatory frameworks.

- **Regulatory Compliance**

Regulatory compliance requires embedding regulations into daily operations rather than treating them as a mere formality. This includes implementing regulatory frameworks issued by SDAIA, such as the Generative Artificial Intelligence Guidelines and Deepfake Guidelines.

B. Enablers

Data

Data forms the foundation of all AI applications, with the quality of outcomes highly dependent on the accuracy, availability, and integration of data. Enablement involves building a robust enterprise structure, enhancing data quality, and ensuring privacy and security, thereby supporting the sustainability and efficiency of AI models. This includes:

- **Availability and Accessibility**

Ensuring that data is available and easily accessible through the use of data discovery tools, data catalogs, and automated updates to maintain currency.

- **Quality and Integration**

Data quality directly affects model accuracy. Therefore, it is essential to track data sources, document usage rights, conduct legal reviews for data usage, and ensure compliance with intellectual property rights.

- **Reliability**

To maintain data continuity and trust, it is recommended to use distributed cloud storage along with backup solutions for disaster recovery.

Infrastructure

Infrastructure involves digital resources such as servers, cloud, and data centers that enable the development and operation of AI applications. It must be flexible, scalable, and secure to ensure consistent performance and accurate results. The related domains include:

- **Technical Standards**

Ensuring compatibility, performance, and sustainability by providing a scalable, flexible infrastructure capable of accommodating growth, as well as integrating with national platforms.

- **Availability and Monitoring**

Focusing on providing a flexible internal infrastructure for high-performance computing that supports AI technologies, offering scalability, operational efficiency, and regular hardware upgrades.

- **Operational Resilience**

Ensuring service continuity by maintaining downtime not exceeding 30 minutes annually and using predictive analytics to reduce potential outages.

Human Capabilities

Human capabilities are the cornerstone of AI adoption, requiring specialized technical expertise in AI design, development, operation, and governance. Human enablement involves attracting top talent, training skilled professionals, building career pathways, and providing a flexible, motivating work environment, in addition to fostering academic partnerships to prepare the next generation of professionals. This includes:

- **Headcount and Diversity**

Effective enterprise adoption requires building diverse teams with deep expertise, aligned with the National Occupational Standard Framework for Data and Artificial Intelligence.

- **Professional Development**

Keeping pace with digital transformation demands a sustainable training plan that supports continuous learning, aligned with job roles, and offering accredited certifications in the field.

- **Academic Collaboration**

Partnering with universities and institutes helps supply the entity with qualified talent and fosters an applied learning environment. It also facilitates attracting talented students through training programs and technology competitions.

- **Job Stability**

In the competitive market for specialized talent, retaining such expertise is a strategic priority. This can be achieved through analyzing employee attrition causes and developing professional and financial incentives, as well as aligning HR policies with the National Occupational Standard Framework.

C. Outputs

Applications

Possessing in-house capabilities to develop AI-based products and applications is one of the most significant indicators of enterprise maturity in this domain.

- **Development and Deployment**

This includes the number of AI applications developed, as well as the development and deployment of intelligent applications that provide tailored solutions to meet the entity's needs through an internal development environment supporting model lifecycle management using frameworks such as MLOps.

- **Privacy and Security**

The entity adopts a "Security and Privacy by Design" approach, integrating encryption and access controls from the earliest stages of analysis. It complies with national regulations, such as the Personal Data Protection Law, and implements immediate recovery plans while conducting periodic risk measurements to ensure the safety and continuity of applications.

- **Operation and Management**

Operation involve tracking model usage and linking performance to enterprise indicators, such as service quality and cost reduction. Intelligent monitoring mechanisms are implemented for proactive improvement, and analytical reports are submitted to senior management to support decision-making and enhance investment in successful models.

Impact

The impact of AI extends from cost reduction and efficiency enhancement to improved user experience and more accurate, effective decision-making. This impact can only be realized when AI technologies are integrated into the enterprise environment in alignment with strategy, governance, and enablers.

- **Operational Efficiency**

AI contributes to improving operational efficiency and productivity by reducing costs through automating manual processes. The impact is measured using return on investment (ROI) models that include productivity gains, time savings, and performance improvements, directing investments toward higher-value solutions. AI supports at least 25% of jobs, with productivity gains targeted at over 15%. Performance analysis tools are used to monitor this improvement and link job training to actual outcomes.

- **Employee Productivity**

AI is employed to analyze beneficiary behavior and personalize services, enhancing user experience and improving time-related performance indicators by over 20%. Interactivity is further strengthened through proactive solutions such as chatbots, with AI integrated across all service channels.

- **Service Quality and Improvement**

This reflects the extent to which the entity adopts AI to deliver interactive, personalized, and proactive services through diverse channels for users, customers, or citizens. AI powers more than 25% of services, with improvements in response time, satisfaction, and personalization exceeding 20%.

Each of these domains is evaluated based on a set of detailed questions and specific criteria outlined in the following sections.

3.2.2 Adoption Measurement Levels

The National AI Index classifies the maturity level of government entities in AI adoption into six main levels. Each level reflects a specific stage in the entity’s practices and development of AI capabilities, and helps define a roadmap for continuous improvement. The levels are as follows:

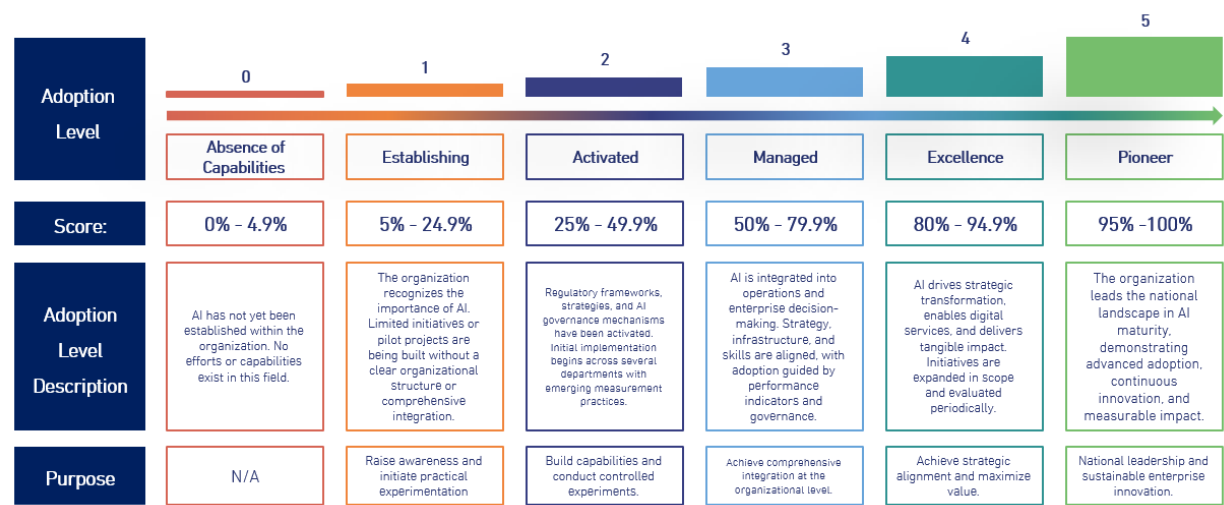


Figure (3): Adoption Measurement Levels

Level 0: Absence of Capabilities

Score: 0% – 4.9%

Description: AI has not yet been established within the organization. No efforts or capabilities exist in this field.

Objective: None.

Level 1: Establishing

Score: 5% – 24.9%

Description: The organization recognizes the importance of AI. Limited initiatives or pilot projects are being built without a clear organizational structure or comprehensive integration.

Objective: Raise awareness and initiate practical experimentation.

Level 2: Activated

Score: 25% – 49.9%

Description: Regulatory frameworks, strategies, and AI governance mechanisms have been activated. Initial implementation begins across several departments with emerging measurement practices.

Objective: Build capabilities and conduct controlled experiments.

Level 3: Managed

Score: 50% – 79.9%

Description: AI is integrated into operations and enterprise decision-making. Strategy, infrastructure, and skills are aligned, with adoption guided by performance indicators and governance.

Objective: Achieve comprehensive integration at the organizational level.

Level 4: Excellence

Score: 80% – 94.9%

Description: AI drives strategic transformation, enables digital services, and delivers tangible impact. Initiatives are expanded in scope and evaluated periodically.

Objective: Achieve strategic alignment and maximize value.

Level 5: Pioneer

Score: 95% – 100%

Description: The organization leads the national landscape in AI maturity, demonstrating advanced adoption, continuous innovation, and measurable impact.

Objective: National leadership and sustainable enterprise innovation.

3.3 Measurement

The percentage of AI adoption by entities is measured in three main steps, beginning with the delivery of introductory workshops, followed by the distribution of questions for completion along with the collection of supporting documents, and concluding with the analysis and verification of responses.

3.3.1 Awareness Workshops

Workshops are conducted with representatives of government entities to raise awareness of the importance of AI adoption, introduce the National AI Index and its objectives, and explain the questions in detail. The workshops also clarify the response mechanism and the process for compiling and sharing supporting documents.

3.3.2 Questionnaire and Supporting Documents

The National AI Index questions are shared with government entities for completion. Entities are also required to compile and share the documents and evidence supporting each response.

3.3.3 Review and Verification

Responses and supporting documents are reviewed and verified for accuracy and validity by a specialized team. Follow-up workshops with entities may be held, where necessary, to address any points requiring clarification.

3.4 Results

This stage represents the final step in implementing the National AI Index. It aims to accurately portray the readiness level of government entities for AI adoption. It relies on data collected during the measurement stage, transforming practices and indicators into quantitative results and benchmark comparisons that enable the identification of gaps and guide future development plans.

3.4.1 Adoption Scores

The adoption score reflects the overall level attained by the entity in utilizing and implementing AI technologies, based on an analysis of its enterprise performance results in this domain. The score is calculated by aggregating and analyzing the results of the sub-domains within the Index components, applying the predetermined weights for each domain. This approach enables the development of a tailored roadmap to raise the level of adoption and determine the entity's position on the maturity scale.

3.4.2 Comprehensive and Detailed Reports

The reporting stage of the National AI Index plays a pivotal role in measuring AI adoption among entities. These reports aim to support decision-makers in government entities by providing an accurate analytical reading of the current state of adoption, enabling entities to set their development priorities based on reliable data.

This stage includes two types of reports:

Comprehensive National Report:

Provides a holistic view of AI adoption levels across the government sector, highlighting areas that have achieved advanced maturity, as well as identifying aspects requiring development or intervention to strengthen enterprise readiness.

Detailed Report for Each Entity:

Based on the measurement results, SDAIA will analyze each entity's performance across various domains and provide targeted recommendations that outline practical steps for improving adoption levels. These recommendations will focus on addressing weaknesses and maximizing the impact of AI utilization. The recommendations aim to:

- Support decision-making within government entities through an objective analysis of the adoption state.
- Enhance the entities' ability to develop future improvement plans with greater accuracy and alignment to national priorities.
- Raise competitiveness and encourage healthy competition among government entities in adopting AI.
- Achieve national alignment in progress toward the objectives of the National Strategy for Data and AI.

This step forms an integral part of the Index's governance framework, contributing to the establishment of a culture of continuous improvement in AI adoption across government entities in Saudi Arabia.

3.4.3 Follow-Up

To enhance the effectiveness of the National AI Index and ensure the sustainability of its impact within the government sector, a comprehensive annual review will be conducted for the Index components, questions, and measurement methodology. This will ensure accuracy, alignment with the sector's actual needs, and responsiveness to the ongoing evolution of AI practices.



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